

THE HISTORY OF THE SOLAR SYSTEM

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THE SOLAR SYSTEM IS THOUGHT to have begun forming about 4.6 billion years ago from a gigantic cloud of gas and dust, called the solar nebula. This cloud contained several times the mass of the present-day Sun. Over millions of years,

it collapsed into a flat, spinning disk, which had a dense, hot central region. The central part of the disk eventually became the Sun, while the planets and everything else in the solar system formed from a portion of the remaining material.

THE FORMATION OF THE SOLAR SYSTEM

No one knows for certain what caused the great cloud of gas and dust, the solar nebula from which the solar system formed, to start collapsing. What is certain is that gravity somehow overcame the forces associated with gas pressure that would otherwise have kept it expanded. As it collapsed, the cloud flattened into a pancake-shaped disk with a bulge at its center. Just as an ice skater spins faster as she pulls in her arms, the disk began to rotate faster and faster as it contracted. The central region also became hotter and denser. In the parts of the disk closest to this hot central region, only rocky particles and metals could remain in solid form. Other materials were vaporized. In due course, these rocky and metallic particles gradually came together to form planetesimals (small bodies of rock, up to several miles in diameter) and eventually the inner rocky planets—Mercury, Venus, Earth, and Mars. In the cooler outer regions of the disk, a similar process occurred, but the solid particles that came together to form planetesimals contained large amounts of various ices, such as water, ammonia, and methane ices, as well as rock. These materials were destined eventually to form the cores of the gas-giant planets—Jupiter, Saturn, Uranus, and Neptune.

6 REMAINING DEBRIS

Radiation from the Sun blew away most of the remaining gas and other unaccreted material in the planetary solar system. Some of the leftover planetesimals in the outer part of the disk formed the vast and remote Oort Cloud of comets.

inner solar system

frozen cometary nuclei

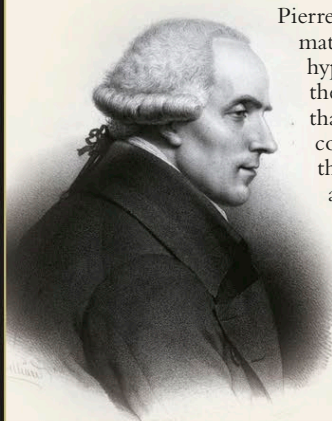
1 SOLAR NEBULA FORMS

The solar nebula started as a huge cloud of cold gas and dust many times larger than our present solar system. Its initial temperature would have been about -382°F (-230°C). From the start, the solar nebula was probably spinning very slowly.

SIX STEPS TO FORM A SOLAR SYSTEM

Shown here is an outline of the nebular hypothesis—the most widely accepted theory for how the solar system formed. It provides a plausible explanation for many of the basic facts about the solar system. For example, it explains why the orbits of most of the planets lie roughly in the same plane and why the planets all orbit in the same direction.

PIERRE-SIMON DE LAPLACE



Pierre Laplace (1749–1827) was a French mathematician who developed the nebular hypothesis—the idea, originally proposed by the German philosopher Immanuel Kant, that the solar system originated from the contraction of a huge gaseous nebula. Today, this hypothesis provides the most widely accepted theory for how the solar system formed. Another of Laplace's contributions to science was to analyze the complex forces of gravitational attraction between the planets. He investigated how these might affect the stability of the solar system and concluded that the system is inherently stable.

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The ring of planetesimals between Mars and Jupiter failed to form a planet, possibly because of the gravitational influence of Jupiter. Instead, they formed a belt of asteroids, including this asteroid, Ida.